

When looking back at my academic journey at Iowa State one thing sticks out to me. It is a wide variety of group projects to help students better learn to work in a team just like the real world. Once I began looking back I started seeing the bigger picture for what these types of classes have enabled for me. Classes like software development pushed me to work with both friends and new people all trying to achieve the same goal. Classes like this have both encouraged and enabled me to improve in things like communication as well as splitting up the work. I am a person who likes to get everything done at once but these large scale team projects required me to let go.

When thinking about what I have learned from my courses about society and global aspects, many of my electives have brought to my attention the importance of the societal impact that engineering has. Iowa State University has requirements for US cultures and communities as well as international perspectives that must have specific electives taken in order to accomplish these requirements. Classes like Liberty and Law in America have brought the importance of community and society to the forefront of my education. The ethics course for cybersecurity that I took also highlights the importance of global aspects of engineering because it taught me about how engineering is a global job and also the global ramifications of mistakes I could make.

Another important aspect of my time at Iowa State is how it has taught me the process of engineering and some of the models of teamwork that that entails. One class COMS 3090 sticks out to me because of its usage of SCRUM for its overarching term project. Another class that had something similar was my CPRE 2880 embedded systems class where students were placed in 4 groups and had to come up with a problem to solve with the CyBot, a Roomba-like robot that could drive around the room. These classes taught me that falling forward is the most important thing you can do as an engineer. The engineering process rewards that because of how iterative it is. For the embedded systems example we had to learn on the fly how our

sensor outputs data in order to be able to tell what the robot is seeing so that our drive team can maneuver the robot around the obstacles in the course.

One of the most important parts of my major was the importance of learning by yourself as you went through problems. One of the most enlightening assignments I had was a project for my operating systems class where we were encouraged to use the internet as well as other resources to assist us in both programming and understanding how operating systems work. Beyond that, the offensive security testing course I took encouraged usage of outside sources to assist in the process of the labs for both bug fixing and question answering. These examples and many other instances of outside assistance have helped me to gain a better understanding of problem solving without always needing another person to answer the questions for you.

For outside activities to enhance my learning I have been a member of undergraduate research all 4 years of my college experience. I am participating in the CyberEd project where our goal is to enhance and create educational materials for younger and older people to help with things like phishing attacks and general cybersecurity awareness. This research experience has enhanced my own learning because it forced me to take the complex terminology and concepts found in a classroom and transition them into digestible and reasonable to understand terminology and that required me to have a deeper understanding of the topics I researched. It also helped me to work in a team because myself and others would collaborate on the material we created to make each piece better than the previous.

Finally, my college experience has left me with knowledge of how to continue my professional journey. Classes like my introduction to security course highlighted many certifications that are beneficial to acquire for someone studying cybersecurity. These certifications grow with time and have shown me that I must do the same. College has taught me that I am on the start of a long journey that requires discipline and growth each and every day. All in all my experience in college has left me with many great memories, opportunities, and lessons that I am forever thankful for.